

Daniel GODINES

PERSONAL DATA

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EDUCATION

JUN 2021 - PRESENT Doctor of Philosophy in ASTRONOMY, **New Mexico State University**, NM
Advisors: Moire PRESCOTT & Wladimir Lyra

AUG 2014 - MAY 2018 Bachelor of Arts in PHYSICS, **Bard College**, NY
Thesis: "A Microlensing Detection Algorithm for Wide-Field Surveys"
Advisor: Antonios KONTOS

PUBLICATIONS

Total Citations: 39 | *h-index*: 3 (Source: NASA/ADS)

Peer-Reviewed Journal Articles

- MAY 2026 [A Hybrid Ensemble and Deep Learning Framework for Detecting High-Redshift Ly \$\alpha\$ Blobs in Broadband Surveys](#)
Godines, D., Prescott, M. K. M., 2026, *PASP*, 138, 054505.
- JAN 2026 [On the Mass Budget Problem of Protoplanetary Disks: Streaming Instability and Optically Thick Emission](#)
Godines, D., Lyra, W., Ricci, L., Yang, C.-C., Simon, J.B., Lim, J., Carrera, D., 2026, *ApJ*, 997, 192.
- OCT 2025 [Dark classification matters: searching for primordial black holes with LSST[†]](#)
Crispim Romão, M., Croon, D., Crossey, B., **Godines, D.**, 2025, *JCAP*, 10, 066.
- SEP 2025 [Anomaly detection to identify transients in LSST time series data[†]](#)
Crispim Romão, M., Croon, D., **Godines, D.**, 2025, *MNRAS*, 543, 351.
- JUN 2019 [A Machine Learning Classifier for Microlensing in Wide-Field Surveys](#)
Godines, D., Bachelet, E., Narayan, G., Street, R. A., 2019, *A&C*, 28, 100298.

[†]Author list is alphabetical.

Research Notes

- SEP 2025 [Classifying Microlensing Events from ROME/REA](#)
Schweitzer, A., Street, R. A., Kruszynska, K., **Godines, D.**, 2025, *RNAAS*, 9, 232.

Selected Manuscripts in Preparation

- 2026 [Accelerating Planet Formation Population Synthesis with Machine Learning Surrogates](#)
Godines, D., et al., In prep.
- 2026 [Early Detection of Stellar and Exotic Microlensing in the Rubin Era](#)
Godines, D., et al., In prep.
- 2026 [Automated Discovery of Ly \$\alpha\$ Blobs in COSMOS](#)
Godines, D., et al., In prep.

AWARDED TELESCOPE TIME

- 2025A **Gemini North Observatory** | PI (27.7 hours, Band 3)
Automated Identification of Lyman-alpha Blobs in the Early Universe
Awarded via General Queue (GN-2025A-Q-320); Instrument: GMOS-N.
- 2024A **Gemini North Observatory** | PI (5.94 hours, Band 2)
Automated Identification of Lyman-alpha Blobs in the Early Universe
Awarded via Fast Turnaround (GN-2024A-FT-210); Instrument: GMOS-N.

PROFESSIONAL SERVICE

- 2025 **Peer Reviewer** | *Nature Astronomy*
- 2025 **Peer Reviewer** | *Journal of Open Source Software (JOSS)*

RESEARCH EXPERIENCE

- 2016 – 2019 **Research Intern** | LAS CUMBRES OBSERVATORY
Gravitational Microlensing Research with Dr. Rachel A. Street
Engineered **MicroLIA**, a machine-learning engine for real-time microlensing detection. Successfully integrated the framework into the **Rubin Observatory** alert stream and the **Fink** and **Pitt-Google** alert brokers.

SELECTED OPEN-SOURCE PROJECTS

- **MicroLIA** – Machine-learning algorithm for real-time transient detection. Adopted by the LSST Microlensing Science Collaboration and designed for live alert-broker deployment (e.g., FINK, ANTARES, Pitt-Google). Modular architecture supporting general time-series classification and synthetic light curve generation. [Docs](#) | [Zenodo](#)
- **pyBIA** – End-to-end pipeline for image-based classification and anomaly detection. Includes segmentation-driven cataloging, morphological feature engineering, unsupervised artifact rejection, and routines for training optimized supervised ML pipelines, including deep-learning models with automated data augmentation. [Docs](#) | [Zenodo](#)
- **protoRT** – Physics-based 3D radiative transfer framework for synthetic radio emission in planet formation simulations. Computes optical depth and emergent intensity through dust density cubes, supporting self-gravitating mono and polydisperse dust with scattering opacity and DSHARP integration. Includes a mass-excess diagnostic for quantifying observational biases from optically thick regions. [Docs](#) | [Zenodo](#)
- **rubin-lc-simulator** – Open-source framework for simulating realistic multi-band LSST light curves, accounting for official cadence strategies and photometric noise. [Docs](#)

PRESENTATIONS

- Contributed Talk**, *Rare Gems in Big Data*, Tucson, AZ, USA 2024
“Classification of Rare Astrophysical Objects: From Galactic to Extragalactic”
- Contributed Talk**, *ODIN Collaboration Meeting*, Seoul, South Korea 2024
“Automated Ly α Blob Detection Using Broadband Imaging”
- Poster**, *50 years of Binaries and Disks: Lubow@75*, Las Vegas, NV, USA 2024
“On the Mass Budget Problem of Planet Formation Theory: Streaming Instability and Optically Thick Regions”

SCHOLARSHIPS AND AWARDS

2025 - 2026 Zia Award for Excellence in Research
2023 - 2024 A. Scott Murrell Memorial Endowed Scholarship
2021 - 2023 William Webber Voyager Graduate Fellowship
2017 - 2018 LSSTC Enabling Science Program Award
2014 - 2018 Distinguished Scientist Scholar Award

SELECTED TECHNICAL SKILLS

Programming Proficiency: PYTHON (Expert), C++, SQL (MySQL), MATLAB, MATHEMATICA, HTML
Data Science & ML: TensorFlow, PyTorch, scikit-learn, XGBoost, emcee, astropy, rubin-sim
Software & Tools: Git, Docker, Sphinx (Documentation), Zenodo, DS9, PHOTOSHOP
Operating Systems: GNU/LINUX (Ubuntu/CentOS), MAC OS X
Language Proficiency: English, Spanish (Native), French (Conversational)

TECHNICAL TRAINING & OBSERVATIONAL EXPERIENCE

SUMMER 2024 **ZTF Summer School** | CALIFORNIA INSTITUTE OF TECHNOLOGY
Advanced training in ZTF data releases and alert-broker integration for transients.

SUMMER 2018 **ZTF Summer School** | CALIFORNIA INSTITUTE OF TECHNOLOGY
Intensive program on wide-field survey analysis and automated discovery.

SUMMER 2018 **Research Visit** | SPACE TELESCOPE SCIENCE INSTITUTE (STScI)
Optimization of machine learning pipelines for high-dimensional astrophysical data.

SUMMER 2017 **CUREA Program** | MT. WILSON OBSERVATORY
Two-week residence in observational stellar and solar physics.

SUMMER 2017 **Technical Workshop** | NATIONAL OPTICAL ASTRONOMY OBSERVATORY (NOAO)
Training on LSST data infrastructure and real-time alert systems.

SUMMER 2016 **WIYN 3.5m Observatory** | KITT PEAK NATIONAL OBSERVATORY
Observer for a joint imaging campaign supporting the *Kepler* K2 mission.

SUMMER 2016 **iPTF Summer School** | CALIFORNIA INSTITUTE OF TECHNOLOGY
Foundational training in Time-Domain Astronomy and survey operations.

LEADERSHIP & OUTREACH

- 2024–PRESENT **Board Member** | SANTA BARBARA COUNTY SCIENCE & ENGINEERING FAIR
Managed logistics and STEM advocacy for grades 6–12 across Santa Barbara and San Luis Obispo counties; coordinating pathways to State finals.
- JUNE 2024 **Invited Lecture** | SANTA BARBARA SALON SERIES
“From Starlight to Artificial Intelligence: How We Learn to Read the Universe” (Public Talk).
- JUNE 2024 **Invited Lecture** | SANTA BARBARA INSTITUTE OF WORLD CULTURE
“Cosmic Revelations: Exploring Astronomy’s Frontiers” (Public Talk).
- 2022 **Guest Science Columnist** | LAS CRUCES SUN-NEWS
Authored features on *A.I. in Astronomy* and *The Search for Planet 9*.
- 2014 **Media Feature** | SANTA BARBARA INDEPENDENT
Profiled in cover story *“A Comet in Santa Barbara’s Astronomical World.”*
- 2013–2014 **Planetary Science Tutor** | SANTA BARBARA CITY COLLEGE
Youngest appointed tutor for undergraduate planetary science course.